ME25102: Basic Thermodynamics

First Law of Thermodynamics

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Course Objectives						
1	To introduce the fundamentals of thermodynamics					
2	To understand the concepts of laws of thermodynamics					
3	To apply the concepts of thermodynamics towards open and closed systems.					
Course Outcomes						
1	**DESCRIBE the basics of thermodynamics with heat and work interactions.**					

Energy-

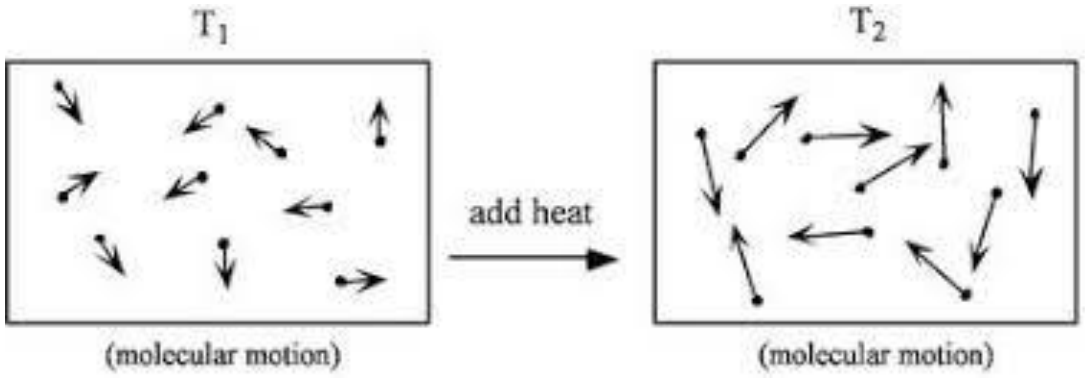
- The capacity to do work
    - Potential Energy, Kinetic Energy, Thermal, Electrical, Mechanical, Magnetic
    - Elastic, Chemical, Nuclear, Light, Sound

- Energy Stored- Chemical, Potential, Nuclear, Kinetic, Potential, Elastic etc

- Energy in Transit- Work and Heat

Heat-

- **Energy due to** the movement of atoms, molecules or ions in solids, liquids and gases.



- Unit- **cal or kcal**
- (Energy required to increase the temperature of 1kg water)
- SI Unit - **Joule (J) or kilo Joule (kJ)**

Work-

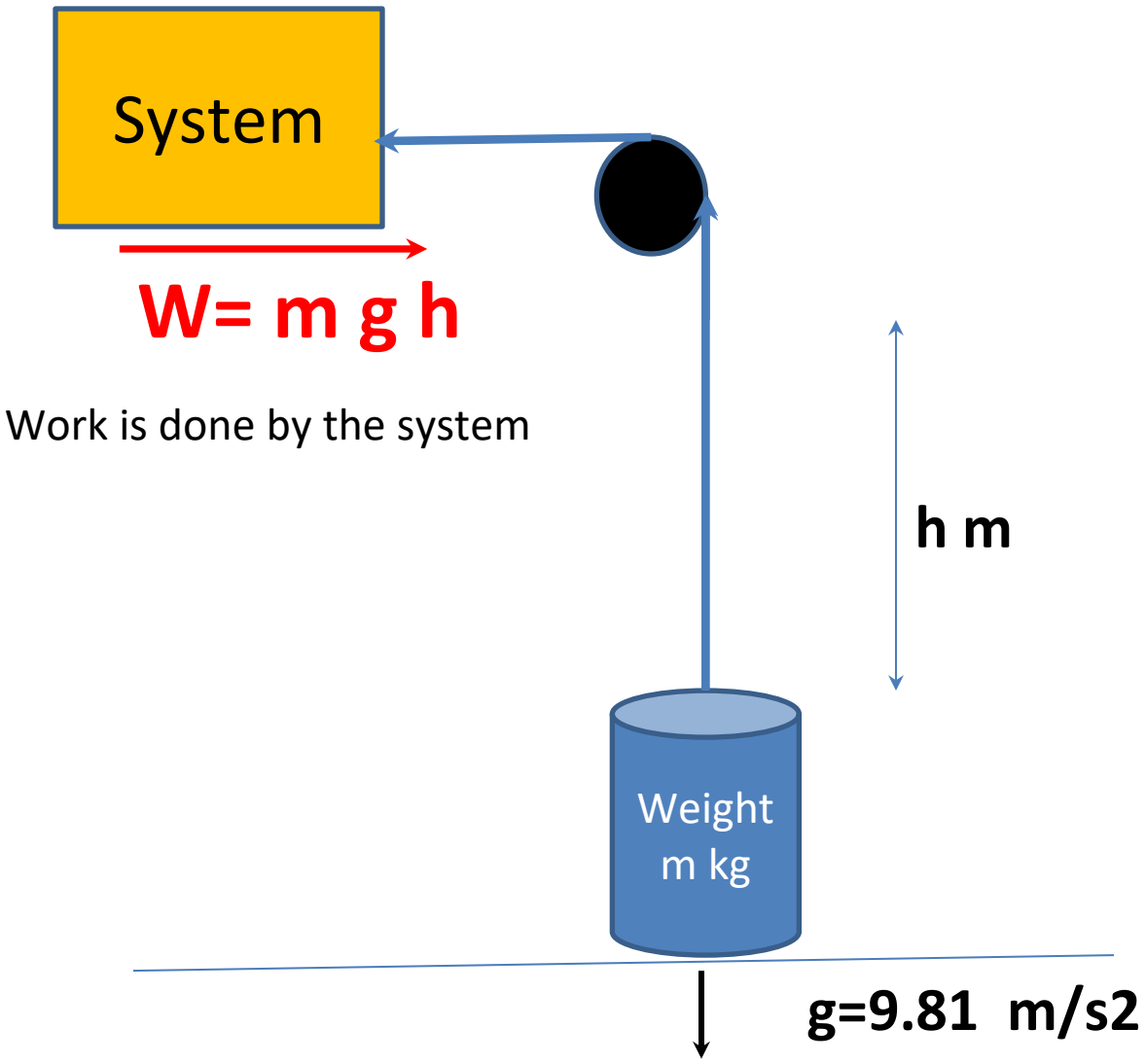
- Definition from physics

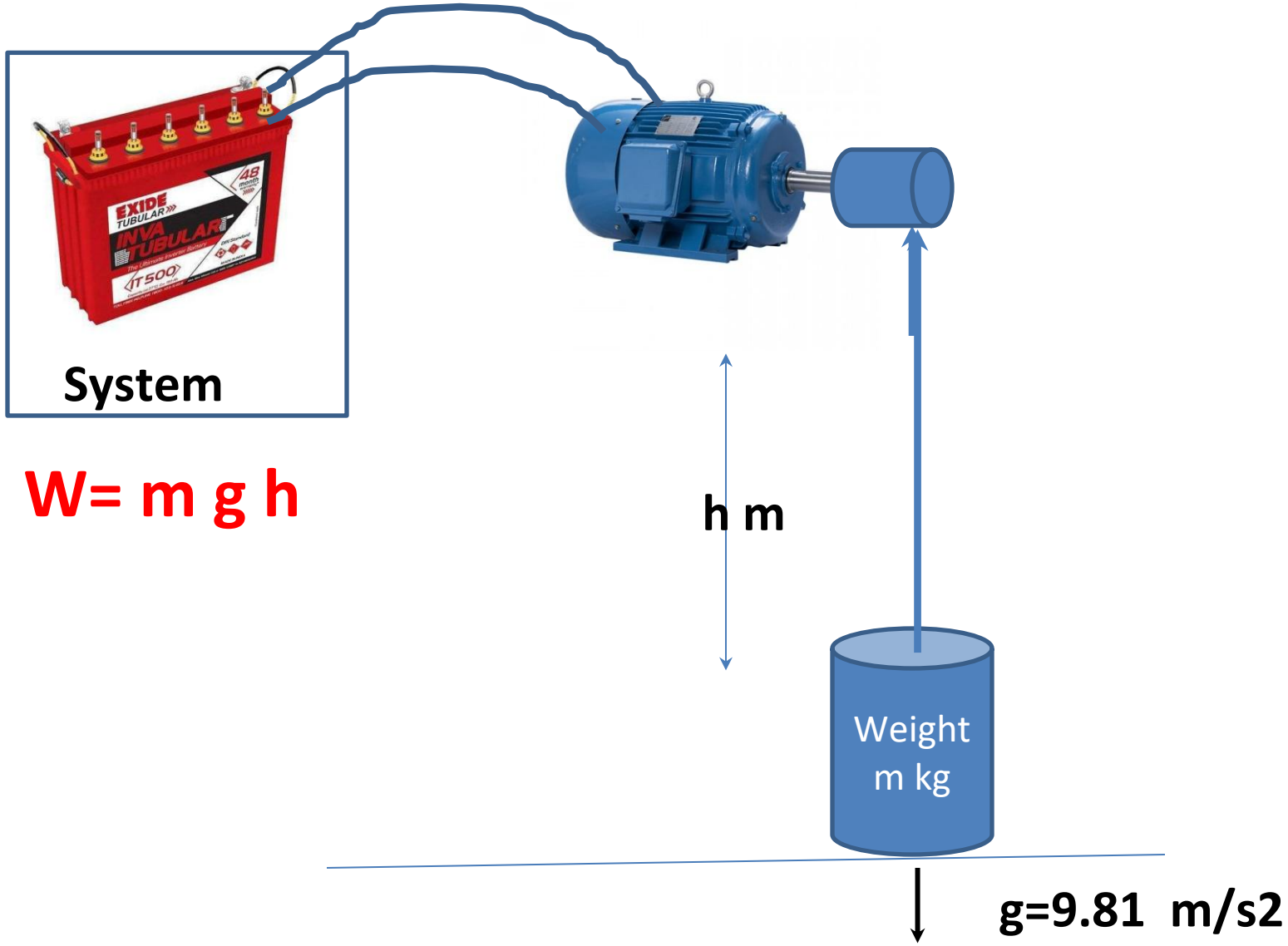
Work= Force X Displacement

W= F X L Joule/ kilo Joule

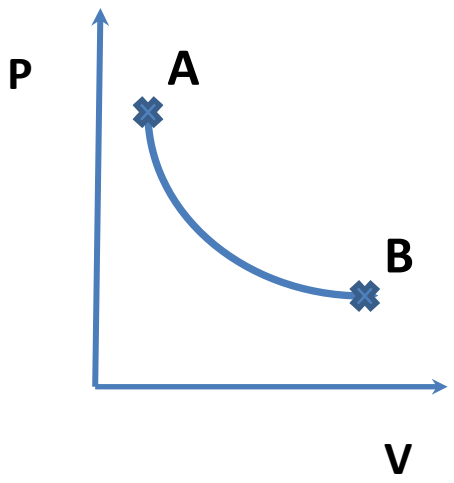
- To define all types of works in one form. Mechanical Work is defined as

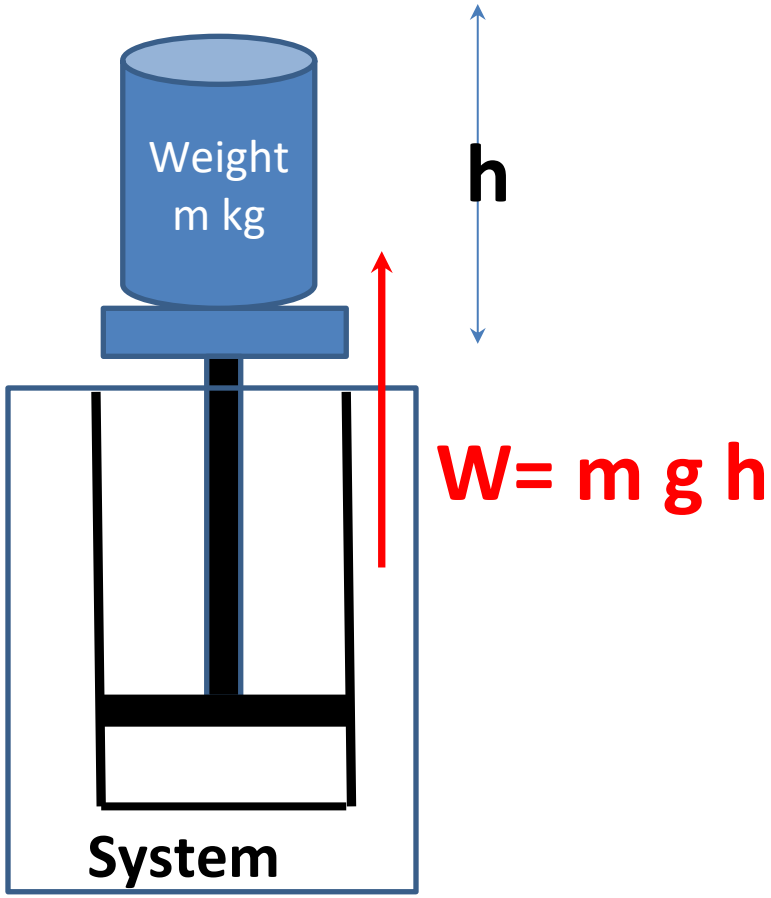
- Work is said to be done by a system if the sole effect of things external to the system can be reduced to raising of a weight


System
W= m g h
Work is done by the system
h m
Weight
m kg
g=9.81 m/s2


EXIDE
TUBULAR
INVA TUBULAR
IT500
System
h m
Weight
m kg
g=9.81 m/s2

W= m g h


P
A
B
V

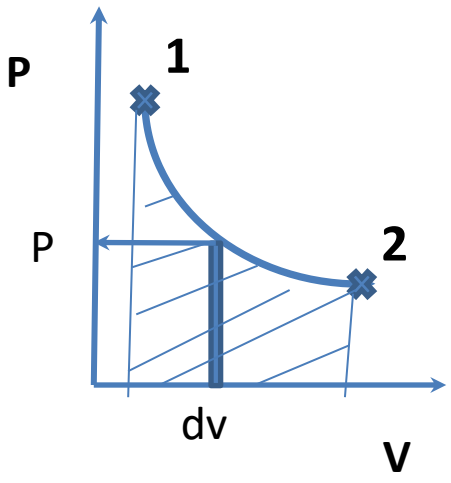

Weight
m kg
h
W= m g h
System

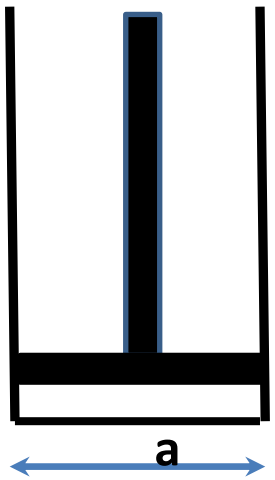
P.dV Work

$$dW = F.dl$$

$$dW = P.a.dl$$

$$dW = P.dv$$



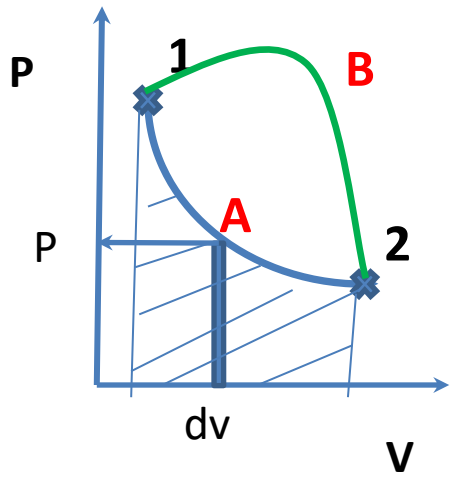


System

$$W = \int_1^2 dW = \int_1^2 P\,dv$$

$$W = \int_1^2 P\,dv$$

Work as Path Function



$$dW = P.dv$$

$$W = \int_1^2 dW = \int_1^2 P\,dv$$

$$W = \int_1^2 P\,dv$$

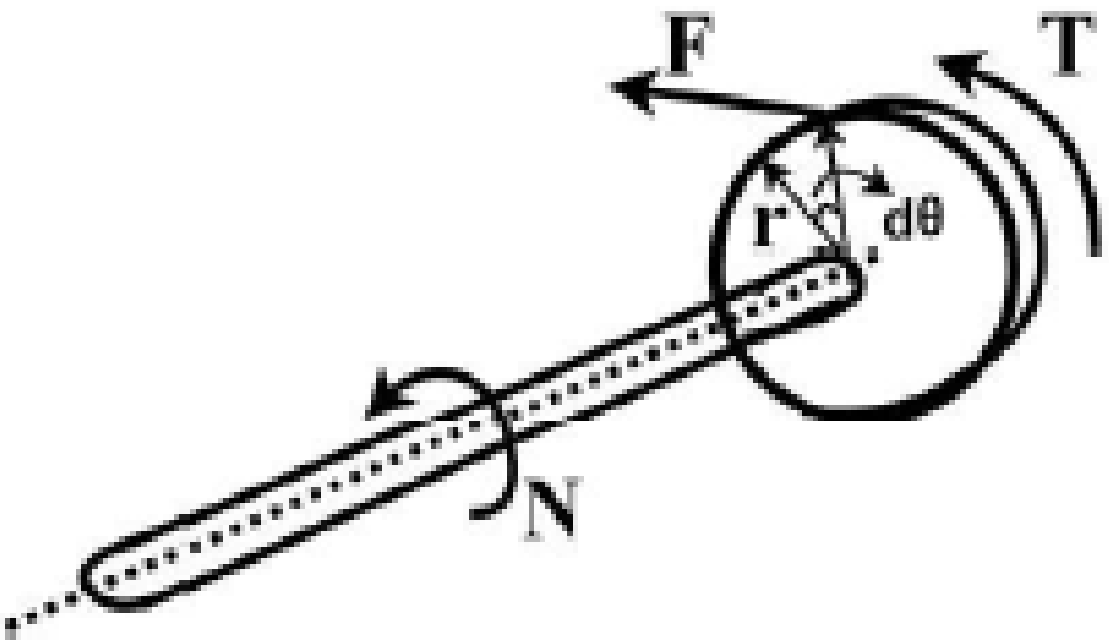
$$W_A =? W_B$$

$$\int_1^2 P.dv_{\;A} =? \int_1^2 P.dv_{\;B}$$

$$W_A \neq W_B$$

Work depends on Path

Shaft Work



$$T = F.r$$

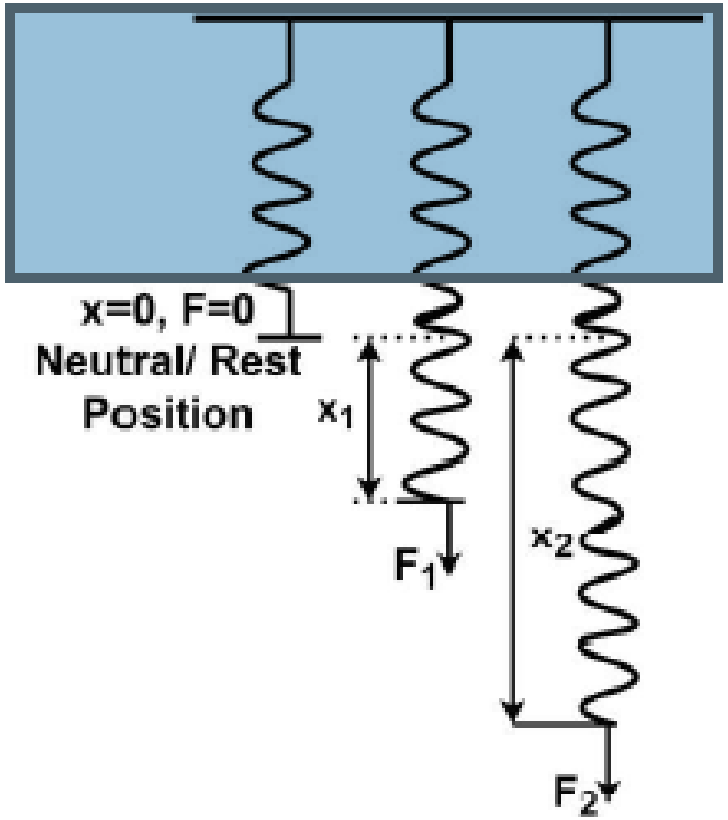
$$dw = F.rd\theta$$

$$dw = T.d\theta$$

$$\dot{W} = \frac{\delta W}{dt} = T\frac{d\theta}{dt} = T\omega$$

Here ω is the angular velocity of the shaft and is given as $\omega = \frac{2\pi N}{60}\ rad/s$ where N is the revolution of the shaft per minute.

Spring Work



$$F = kx$$

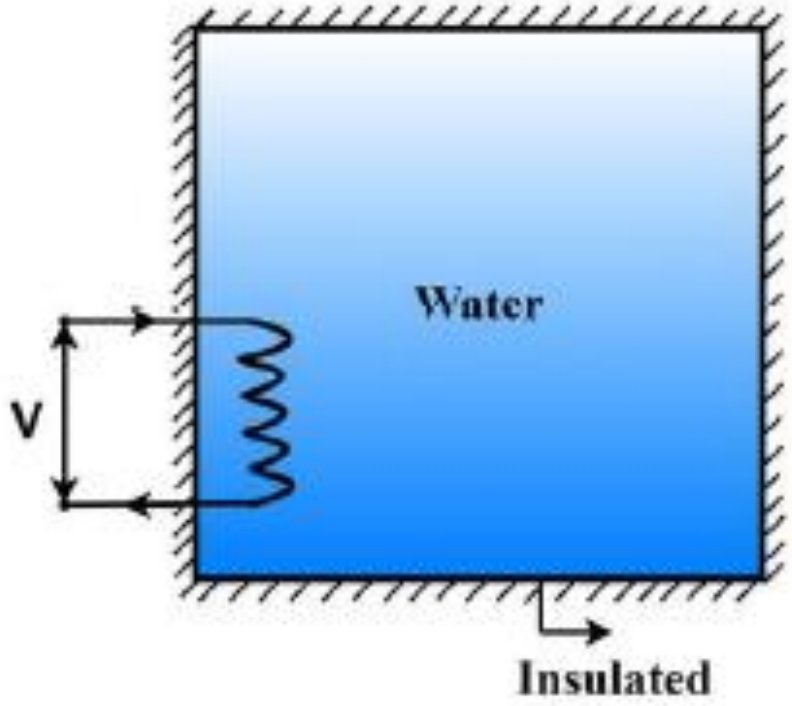
where is the spring constant and its unit is N/m

$$W = \int_{X1}^{X2} F.dx$$

$$W = \int_{X1}^{X2} kx.dx$$

$$W = \frac{1}{2}k(X2^2 - X1^2)$$

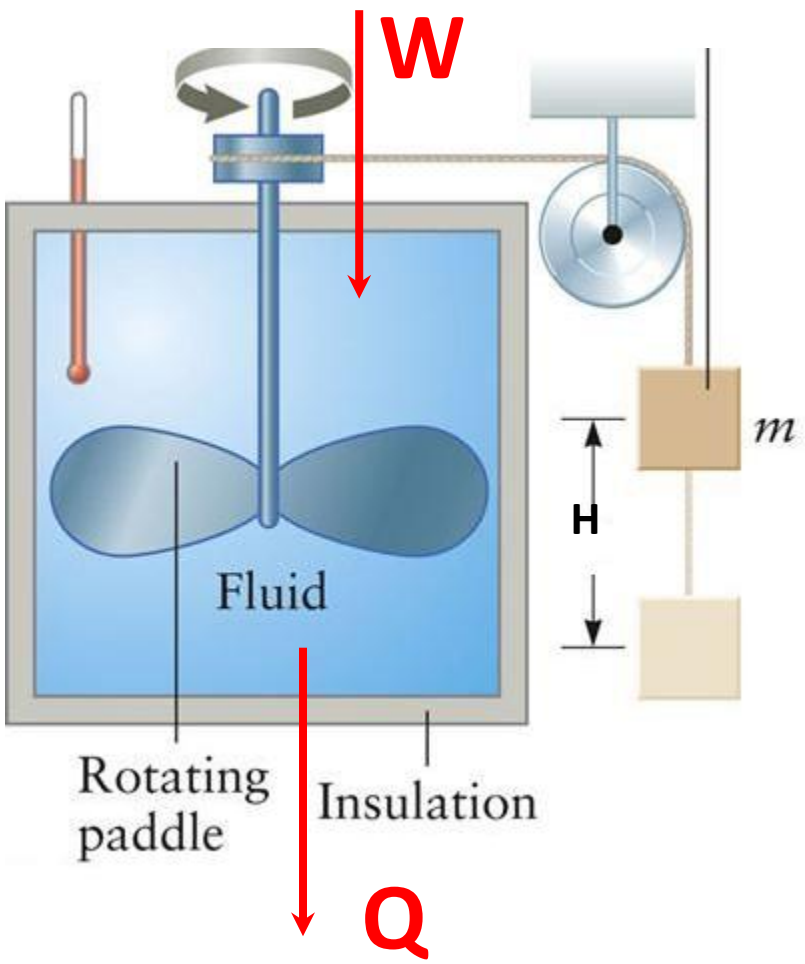
Electrical work

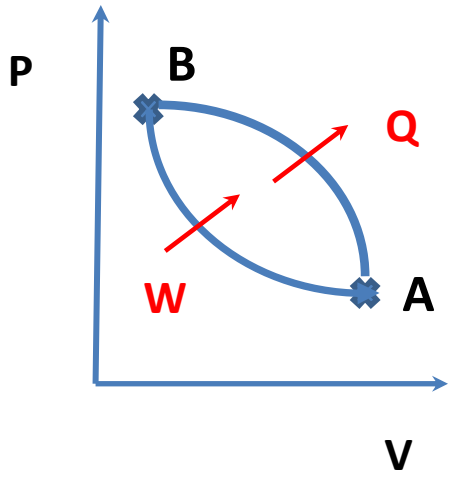


$$We = V.I$$

Here, We is called the electrical power,
the number of electrical charges moving per unit time is electric current (I) and its unit is ampere (A).

Joules Experiment





$Q \alpha W$

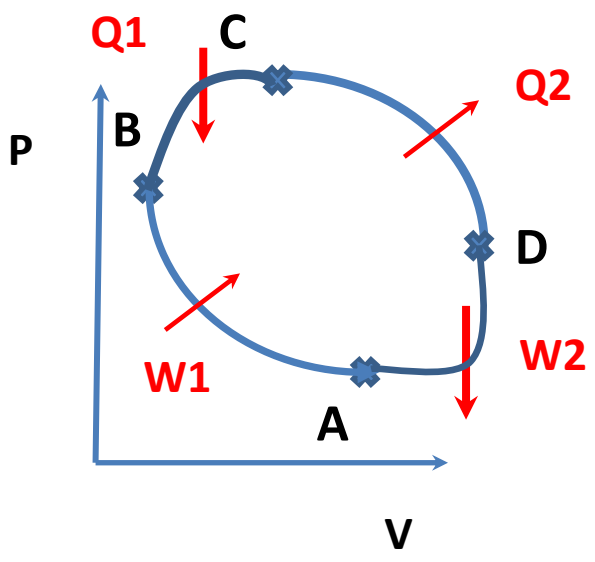
$\Sigma Q = J \Sigma W$

$Q = m\, Cpw\, \Delta T$

$W = m\, g\, H$

$J = 1\ kJ/kJ$

First Law for cyclic processes



$$Q \ \alpha \ W$$

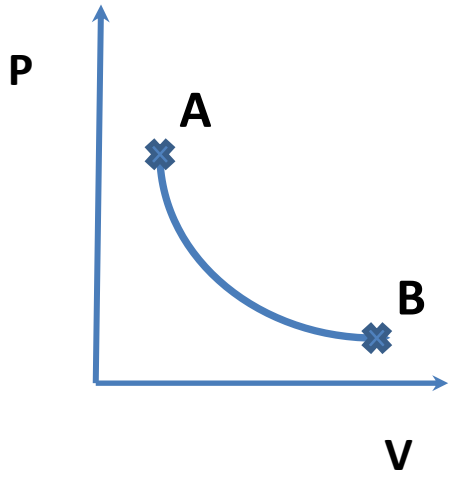
$$(Q1-Q2) = (W2-W1)$$

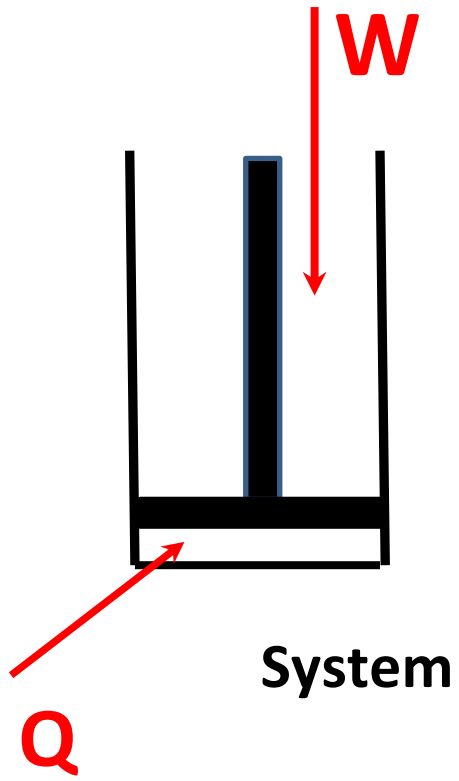
$$\Sigma Q = \Sigma W$$

$$\oint Q = \oint W$$

First Law statement for closed system and cyclic processes

First Law statement for closed system and non cyclic processes





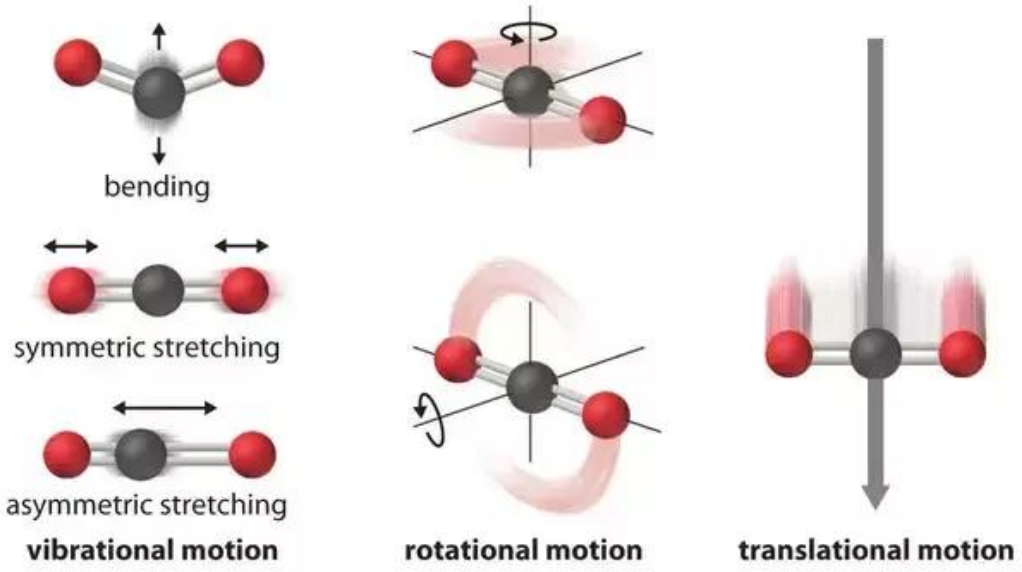
$$Q = W + \Delta E$$

E = Macroscopic Energy + Microscopic Energy

- Macroscopic Energy= K.E.+ P.E. + Magnetic Energy

- Microscopic Energy= Energy related to molecules

- Microscopic Energy= Vibrational + Molecular KE+ Translational Energy + Rotational Energy



$$Q = W + \Delta E$$

E = Microscopic Energy + Macroscopic Energy

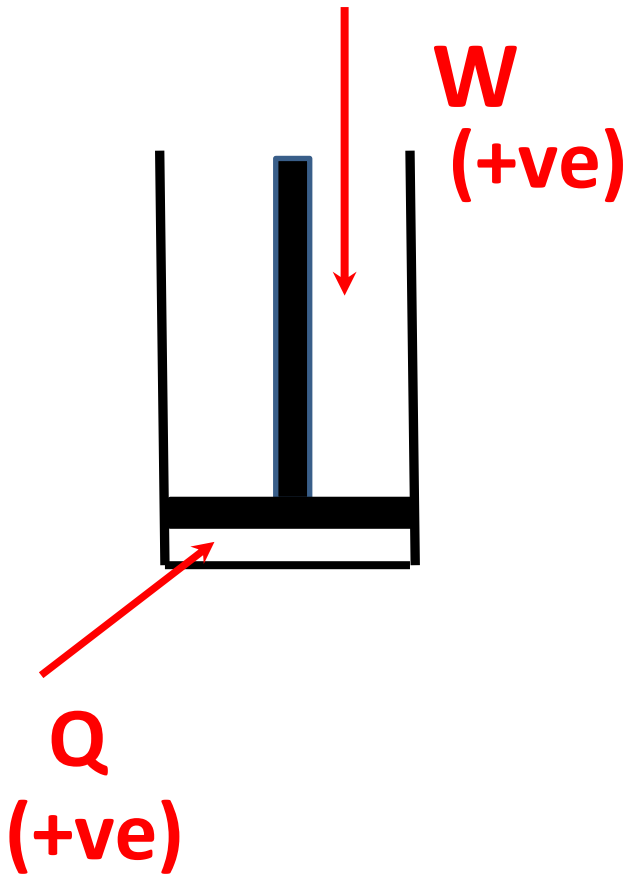
E = Microscopic Energy = Internal Energy (U)

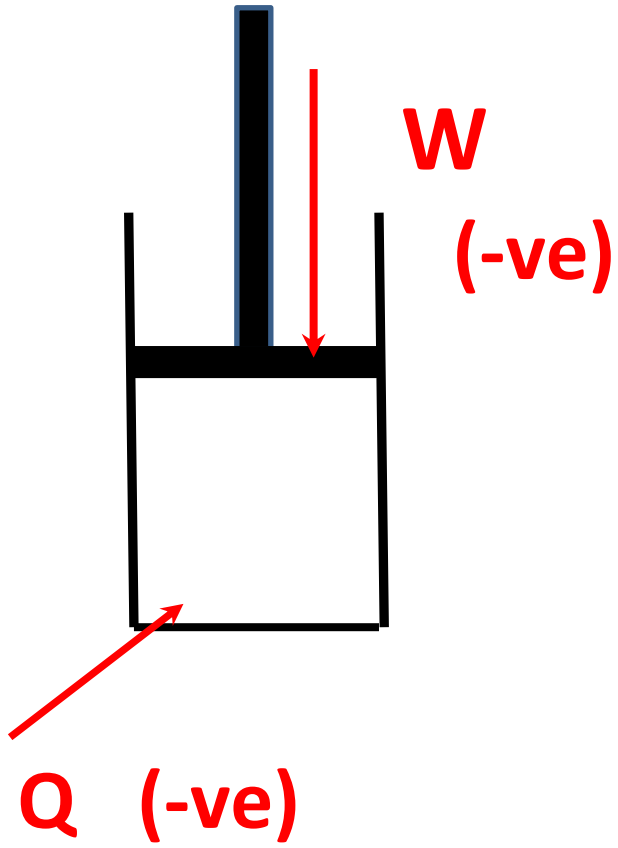
$$Q = W + \Delta U$$

First Law statement for closed system and non cyclic processes

$$Q = W + \Delta U$$

Sign Convention



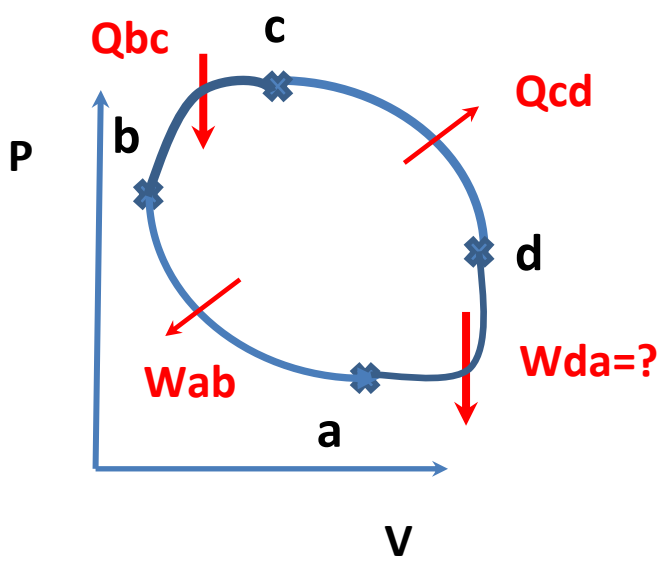


Q time?

https://classroom.google.com/c/MTM2MzQxMTYxODMw

•A piston and cylinder machine contains a fluid system which passes. During a cycle, the sum of all heat transfer is -170 kJ. The system completes 100 cycles per min. Complete the following table showing the method for each item and computes the net rate of work output in kW.

Process	Q (kJ/min)	W (kJ/min)	ΔU (kJ/min)
a-b	0	2170	-
b-c	21000	0	-
c-d	-2100	-	-36600
d-a	-	-	-



By using First law, for closed system and noncyclic processes

- For processes ab,

$$Q_{ab} = W_{ab} + \Delta U_{ab}$$

$$0 = 2170 + \Delta U_{ab}$$

$$\Delta U_{ab} = -2170 \text{ kJ/min}$$

- For processes bc,

$$Q_{bc} = W_{bc} + \Delta U_{bc}$$

$$21000 = 0 + \Delta U_{bc}$$

$$\Delta U_{bc} = 21000 \text{ kJ/min}$$

- For processes cd,

$$Q_{cd} = W_{cd} + \Delta U_{cd}$$

$$-2100 = W_{cd} - 36600$$

$$W_{cd} = 34500 \text{ kJ/min}$$

- For processes da,

$$Q_{da} = W_{da} + \Delta U_{da}$$

$$\Delta U_{da} = Q_{da} - W_{da}$$

By using First law, for closed system and cyclic processes

For abcd cycle

$$\oint Q = \oint W$$

$$\Sigma Q = \Sigma W$$

$$Q_{ab}+Q_{bc}+Q_{cd}+Q_{da} = W_{ab}+W_{bc}+W_{cd}+W_{da}$$

$$0+21000-2100+Q_{da} = 2170+0+34500+W_{da}$$

$$Q_{da}-W_{da} = 17770 \text{ kJ/min}$$

$$\Delta U_{da} = 17770 \text{ kJ/min}$$

Given: **During a cycle, the sum of all heat transfer is -170 kJ/cycle**

$$\Sigma Q = -170 \times 100 = -17000 \text{kJ/min}$$

$$Q_{ab}+Q_{bc}+Q_{cd}+Q_{da}= -17000$$

$$0+21000-2100+Q_{da}= -17000$$

$$Q_{da}= -35900 \text{ kJ/min}$$

- For processes da,

$$Q_{da} = W_{da} + \Delta U_{da}$$

$$W_{da}=-35900-17770= -53670 \text{ kJ/min}$$

Process	Q (kJ/min)	W (kJ/min)	ΔU (kJ/min)
a-b	0	2170	-2170
b-c	21000	0	21000
c-d	-2100	34500	-36600
d-a	-35900	-53670	17770
Σ	-17000	-17000	0

The net rate of work output in kW=?

The net rate of work output in kW=ΣW in kW

The net rate of work output in kW= -17000/60 kJ/s

The net rate of work output in kW= -283.33 kJ/s or kW

I. First Law statement for closed system and cyclic processes

$$\oint Q = \oint W$$

II. First Law statement for closed system and non cyclic processes

$$Q = W + \Delta U$$

1. A water heater of 5kW power is used to heat 10 Litres of water for 10 minutes. If the specific heat of water is 4.18 kJ/kg K , determine the rise in its temperature.
2. Piston cylinder system is used to compress a gas to one half of its original volume and a heat of 200kJ is taken out of the gas during this process while its internal energy remains the same. How much amount of work is done on the gas?

Specific Quantity

= Total Quantity pre unit mass

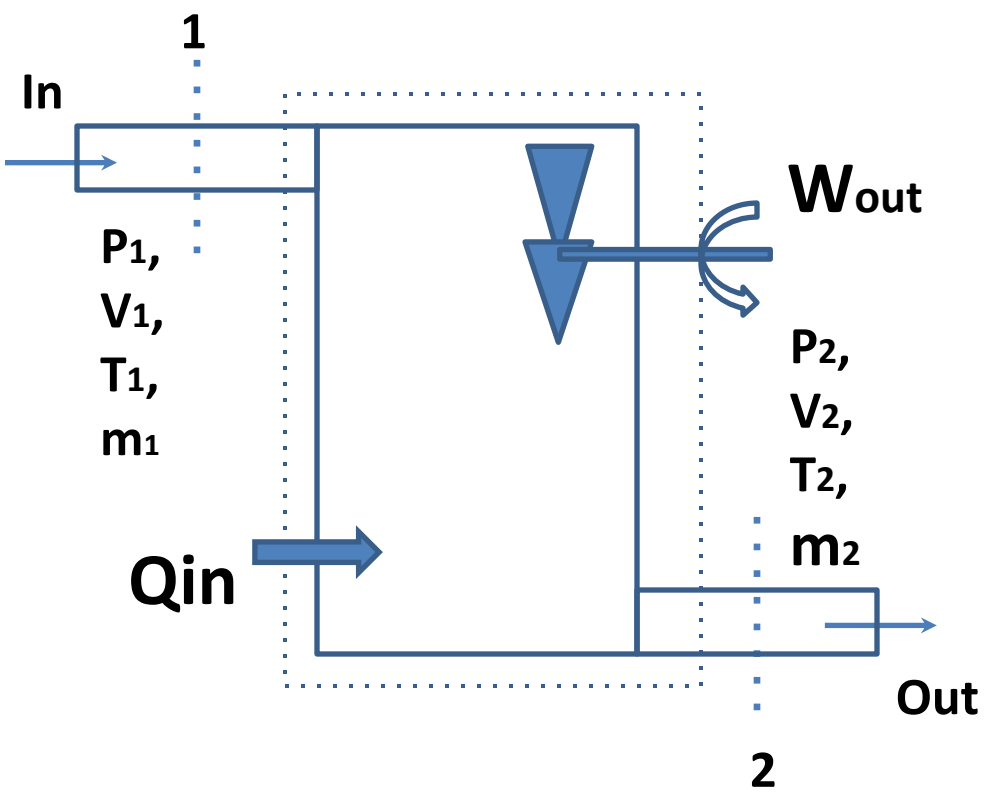
Sr No	Total Quantity	Unit	Specific Quantity	Unit
1.	**Total Volume (V)**	**m^3**	Specific Volume (*v*)	m^3 **/kg**
2.	**Total Energy (E)**	**kJ**	Specific Energy (e)	kJ **/kg**
3.	**Total Internal Energy (U)**	**kJ**	Specific Internal Energy (u)	kJ/kg

A piston-cylinder assembly contains 4 kg of steam. The specific internal energy of the steam changes from 2700 kJkg to 2650 kJkg when it expands from state 1 to state 2. During this expansion, 80 kJ of heat is transferred to the steam and a paddle wheel transfers 18. 5 kJ of work to the steam. Find work done by the steam on the piston.

A gas of mass 10 kg expands within a flexible container so that the p−V relationship is of the form pV1.2= constant. The initial pressure is 1000 kPa and the initial volume is 1 m3 . The final pressure is 10 kPa. If the specific internal energy of the gas decreases by 100 kJ/kg, the magnitude of heat transfer is _____ kJ.

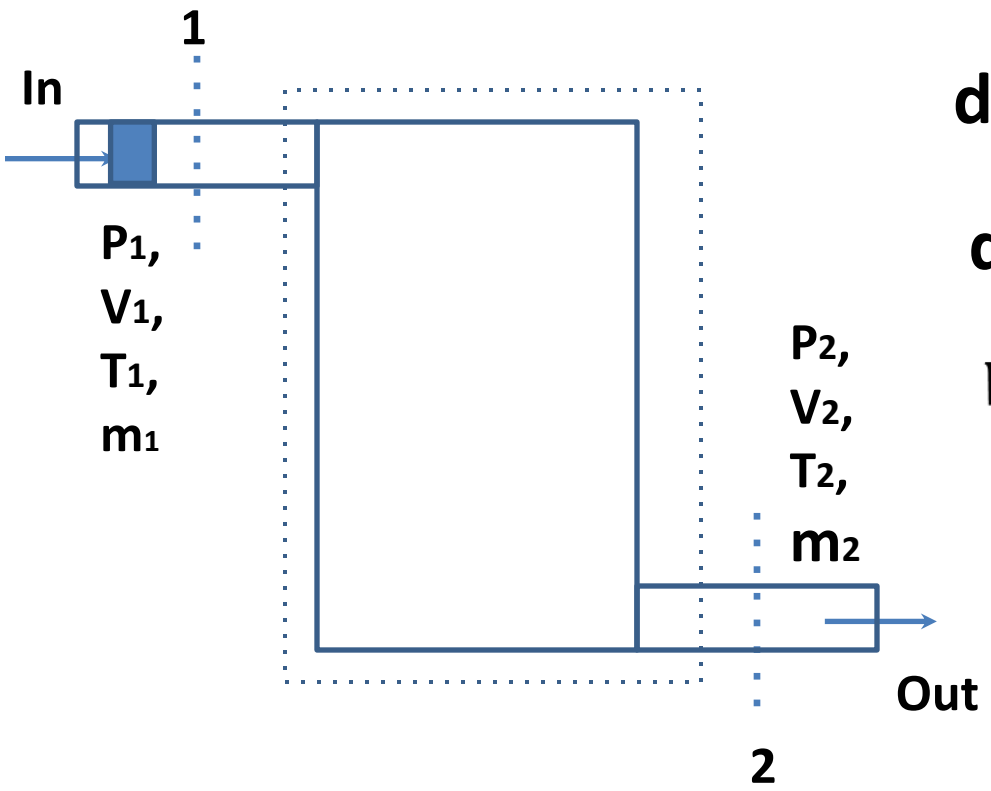
5 kg of air is contained in a vertical piston cylinder assembly with the top open to the atmosphere. The piston weighs 50 kg and has a face area of 0.01 m^2 . The air initially occupies a volume of 0.005 m^3 . The air now undergoes a process wherein its volume decreases to 0.0025 m 3 and 1.5 *kJ* of heat is lost to the surroundings. Find the change in the specific internal energy of the air. Take the atmospheric pressure to be 101325 Pa and g = 9. 81 m/s^2

III. First Law statement for opened system



Turbine, Compressor, Pump etc

Flow Work



$$dW = P.dV$$

$$dW = P.v.dm$$

$$dW = P.v.dm$$

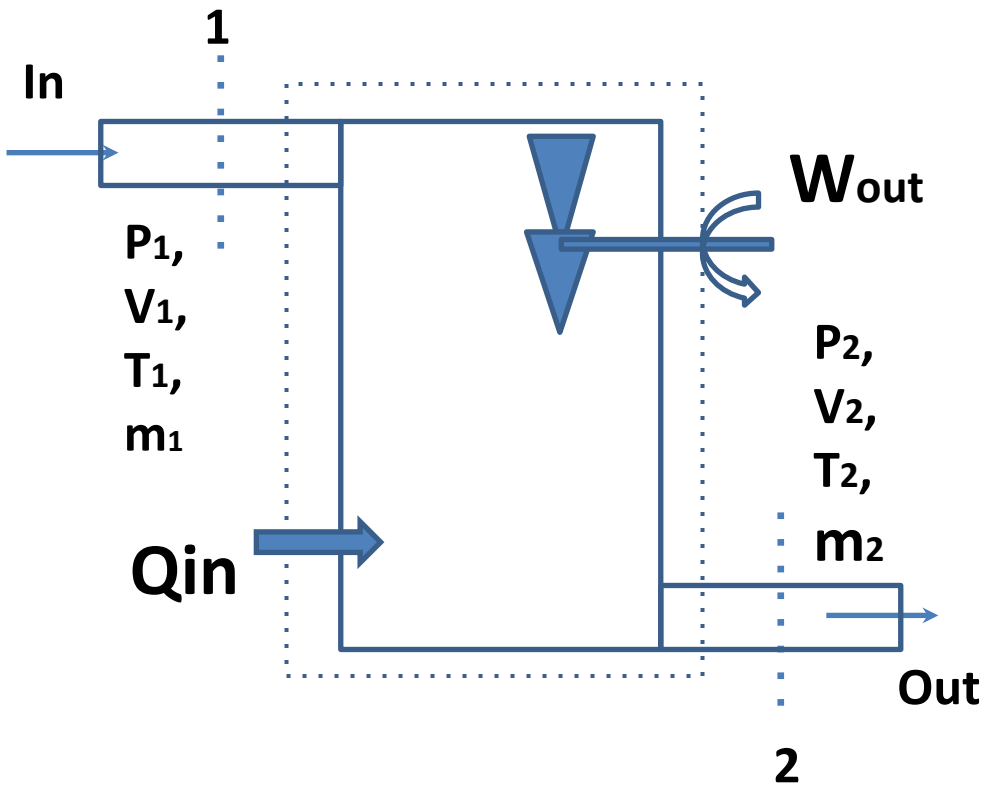
$$W = \int P\, v\, dm$$

$$W_{flow} = P.V$$

$$W_{flow1} = P_1.V_1$$

III. First Law statement for opened system

Law of conservation of energy



Total Inlet Energy = Total Outlet Energy

Total Inlet Energy = Total Outlet Energy

Flow Energy$_1$ + Internal Energy$_1$ + P.E.$_1$ + K.E.$_1$ + Heat Transfer

=

Flow Energy$_2$ + Internal Energy$_2$ + P.E.$_2$ + K.E.$_2$ + Work Out

$$p_1.v_1 + u_1 + z_1 + \frac{1}{2} v_1^2 + q_{in} = p_2.v_2 + u_2 + z_2 + \frac{1}{2} v_2^2 + w_{out}$$

$$\underbrace{p_1.v_1 + u_1}_{h_1} + z_1 + \frac{1}{2} v_1^2 + q_{in} = \underbrace{p_2.v_2 + u_2}_{h_2} + z_2 + \frac{1}{2} v_2^2 + w_{out}$$

$$h_1 + z_1 + \frac{1}{2} v_1^2 + q_{in} = h_2 + z_2 + \frac{1}{2} v_2^2 + w_{out}$$

$$p_1.v_1 + u_1 + \frac{1}{2} v_1^2 + z_1 + q_{in} = p_2.v_2 + u_2 + \frac{1}{2} v_2^2 + z_2 + w_{out}$$

$$h1 + \frac{1}{2} v_1^2 + z_1 + q_{in} = h2 + \frac{1}{2} v_2^2 + z_2 + w_{out}$$

First Law statement for opened system

Steady flow energy equation (SFEE)

Enthalpy (H or h)

$$H = U + PV$$

kJ

$$h = u + Pv$$

kJ/kg

Specific Heat (C)

- Amount of heat required to increase unit temperature (℃or K) of unit mass (kg) of the substance

$$C = \frac{dQ}{m.\Delta T} \qquad {}^{kJ}/_{kg.K} \quad 0r \quad {}^{kJ}/_{kg.℃}$$

$$C = \left(\frac{dq}{dT}\right)$$

$$dq = C.dT$$

Cv

$$Cv = \left(\frac{du}{dT}\right)_{V=C}$$

$$du = (Cv\ .dT)_{V=C}$$

$$dU = m.(Cv\ .dT)_{V=C}$$

Cp

$$Cp = \left(\frac{dh}{dT}\right)_{P=C}$$

$$dh = (Cp\ .dT)_{P=C}$$

$$dH = m.\ (Cp\ .dT)_{P=C}$$

$$Q = \Delta U + W$$

$$dq = du + dw$$

$$dq = (du + Pdv)$$

For constant Volume

$$dq_v = (du + Pdv)_{v=c}$$

$$Pdv=0$$

$$dq_v = (du)_{v=c} \quad \ldots\ldots 1$$

$$dq = C.dT$$

$$dq_v = Cv.dT \quad \ldots\ldots 2$$

$$Cv = \left(\frac{du}{dT}\right)_{V=C}$$

$$Qv = \int m.dq = \int m.Cv.dT \qquad = m.Cv.\Delta T$$

$H = U + PV$ $\quad$ $h = u + Pv$

$$dh = du + v.dP + Pdv$$

$$dh = (du + Pdv) + vdP$$

For constant Pressure $\quad$ $V\,dP = 0$

$$dh_p = (du + Pdv)_{p=c} \qquad dh_p = (dq)_{p=c} \quad \ldots\ldots 1$$

$$dq = C.dT \qquad dq_p = Cp.dT \quad \ldots\ldots 2$$

$$Cp = \left(\frac{dh}{dT}\right)_{p=C}$$

$$Qp = \int m.dq = \int m.Cp.dT \qquad = m.Cp.\Delta T$$

Qv=ΔU

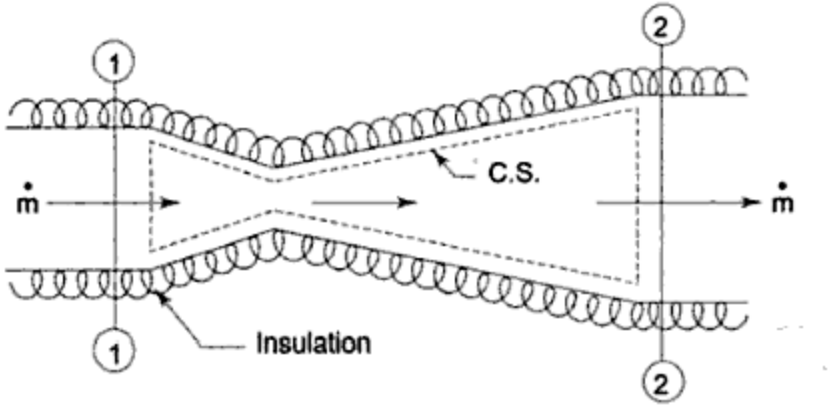
Heat transfer at constant volume=ΔU =$m.Cv.\Delta T$

Qp=ΔH

Heat transfer at constant Pressure=$\Delta H = \Delta U + \Delta PV$

Examples of Steady flow Processes

1. Nozzle



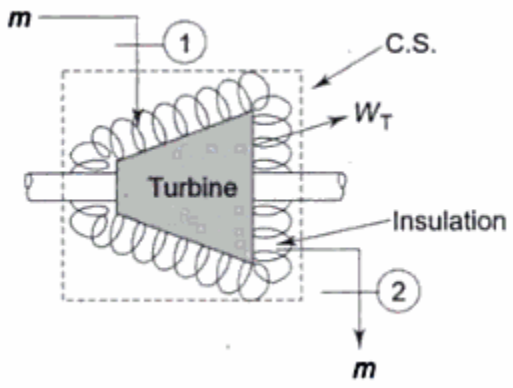
$$h_1 + \tfrac{1}{2} v_1^2 + z_1 + q_{in} = h_2 + \tfrac{1}{2} v_2^2 + z_2 + W_{out}$$

$q_{in} = 0$ $\quad$ $W_{out} = 0$ $\quad$ $z_1 = z_2$ $\quad$ $v_1 <<< v_2$

$$v_2 = \sqrt{2(h_1 - h_2)}$$

Examples of Steady flow Processes

2. Turbine or Compressor



$$h_1 + \frac{1}{2} v_1^2 + z_1 + q_{in} = h_2 + \frac{1}{2} v_2^2 + z_2 + w_{out}$$

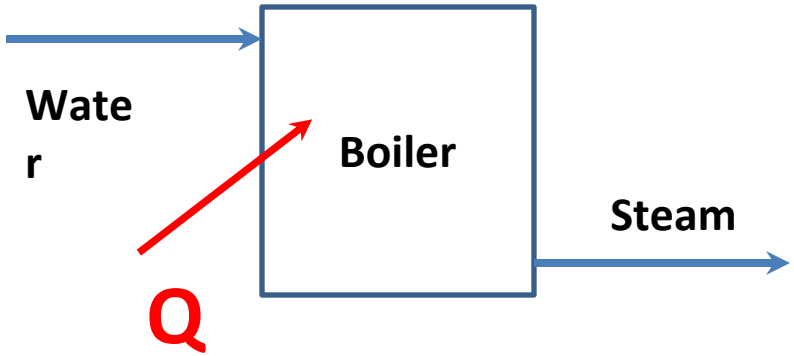
$q_{in} = 0$ $\quad z_1 = z_2$ $\quad v_1 = v_2$

$$w_{out} = (h_1 - h_2)$$

$$W_{out} = m.(h_1 - h_2)$$

Examples of Steady flow Processes

3. Heat Exchanger /Boiler



$$h_1 + \tfrac{1}{2} v_1^2 + z_1 + q_{in} = h_2 + \tfrac{1}{2} v_2^2 + z_2 + w_{out}$$

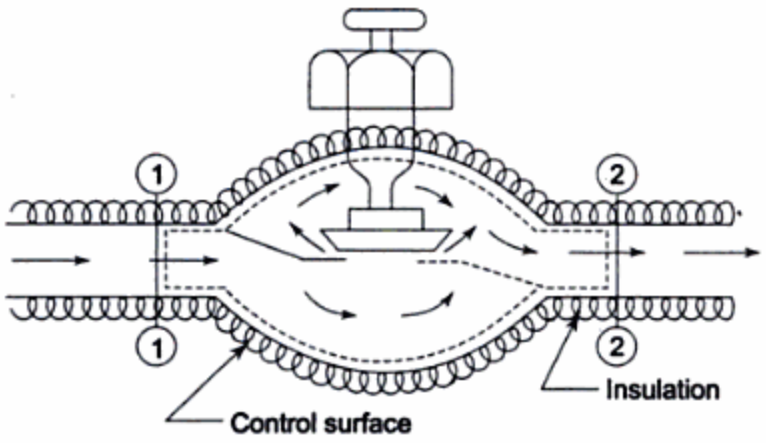
$$W_{out} = 0 \qquad z_1 = z_2 \qquad v_1 = v_2$$

$$q_{in} = (h_1 - h_2)$$

$$Q = m.(h_1 - h_2)$$

Examples of Steady flow Processes

4. Throttling Device



$$h_1 + \frac{1}{2} v_1^2 + z_1 + q_{in} = h_2 + \frac{1}{2} v_2^2 + z_2 + w_{out}$$

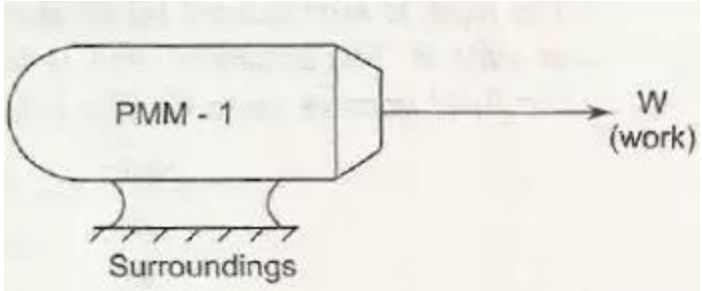
$q_{in} = 0$ $\quad$ $w_{out} = 0$ $\quad$ $z_1 = z_2$ $\quad$ $v_1 = v_2$

$$h_1 = h_2$$

Isenthalpic Processes

- **Perpetual motion machine of First kind (PMM-I)**

The imaginary machine which violets first law of thermodynamics



The imaginary machine which produce the energy (work)

Limitation of First law

1. It does not gives " How much energy conversion is possible?"

$$W \rightarrow Q \qquad =100\%$$

$$Q \rightarrow W \qquad <100\%$$

2. It does not states " The Processes is possible or not"

3. It does not states “ The direction of energy flow”